

CameraLa: A Local-First Browser-Native Framework for Remote Psychophysiological Task Sensing

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Abstract – Continuous physiological sensing and synchronized task measurement in naturalistic environments is challenging. Systems relying on cloud-based video transmission face privacy constraints, while uncontrolled conditions—such as variable lighting, posture, motion, and camera placement—can severely degrade measurement quality and cross-environment comparability. We present CameraLa, a privacy-conscious CameraLa is a local-first, browser-native sensing framework designed to address these challenges. By utilizing WebAssembly (Wasm) and WebGL backends, the framework extracts remote photoplethysmography (rPPG) and facial landmarks entirely on the client device, avoiding raw video transmission. Furthermore, it integrates sensing, psychophysical task timing, and local data persistence within a unified client-side runtime. To evaluate the framework, we conducted an exploratory 10-session longitudinal deployment (framework for remote psychological and psychophysiological task experiments). It synchronizes task events with face-image-derived feature records and exports local CSV/ZIP files without transmitting or storing raw facial video. In an exploratory N=1 ~~)~~ across author-participant deployment, ten sessions were conducted in laboratory and home settings, yielding 600 trials in total. After applying a reaction-time criterion of 100–1500 ms, 541/600 trials remained for descriptive analysis. The exported logs contained timestamped task events and reproducible face-image-derived fields. Differences between laboratory and home exported records are reported as deployment-context differences in face-derived and sensor-derived records, not as evidence of psychological, physiological, cognitive, motivational, or rPPG-accuracy effects. The deployment showed that the system could operate across diverse physical contexts and that between-environment differences were observable in the resulting physiological signals. These results suggest that browser-native edge architectures can support deployment-oriented, privacy-conscious physiological sensing across heterogeneous home and laboratory environments.

Keywords : WebAssembly Browser-Native Sensing, Remote Photoplethysmography, Browser-Native Sensing, Privacy-Preserving Frameworks, In-the-Wild Deployment FaceMesh, Local-First Logging, Task-Aligned Measurement, Laboratory and Home Settings

1. Introduction

Continuous monitoring of physiological signals and task-derived metrics in naturalistic, “in-the-wild” environments provides critical insights for ubiquitous computing and human-computer interaction (HCI). While medical-grade sensors (e.g., electrocardiograms) succeed in controlled laboratories, deploying sensor systems in domestic environments introduces significant engineering challenges. The intended use case of CameraLa is a browser-based remote psychological/psychophysiological task experiment conducted in laboratory and home settings. In this use case, the participant sits in front of

a laptop, views task stimuli, and responds by key press. CameraLa is not intended for continuous physiological sensing during ordinary daily activities. Rather, CameraLa supports task-synchronized acquisition of face-derived records while reducing the need to transmit or retain privacy-sensitive raw facial video. Even in this task-based experiment, deployment outside a controlled laboratory introduces practical difficulties for human-interface research, including heterogeneous illumination, camera placement, posture, browser timing, and local data handling.

First, there are strict privacy constraints. In many practical pipelines, Many video-based physiological analysis relies on external processing measurement pipelines rely on raw video recording

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or server-side components. This approach raises privacy concerns, particularly processing. Such approaches can be difficult to deploy in private spaces like bedrooms or living rooms, which can limit user acceptance and longitudinal deployment. Second, heterogeneous physical environments introduce substantial variability in measurement quality. Uncontrolled lighting, variable camera placements, and unconstrained user postures routinely introduce measurement artifacts that corrupt signal extraction. Consequently, deploying sensors in the wild requires studying not just the user, but how the physical environment interacts with the sensor pipeline because raw facial video is privacy-sensitive and may be identifiable. A local-first browser architecture offers an architectural privacy advantage: raw frames can be processed on the client device, while only extracted features, task events, and quality indicators are stored. This does not constitute a dedicated privacy-preserving algorithm; rather, it is a system-design choice that reduces the need to transmit or retain raw video.

Real-time processing is not treated as the only technically possible architecture. Local video recording followed by offline processing could preserve timestamps if implemented carefully. However, the intended use case of Camerale is a browser-based remote psychological/psychophysiological task experiment conducted in laboratory and home settings, where retaining identifiable facial video for later analysis can itself be a privacy and acceptability burden. Camerale therefore extracts face-image-derived records during task execution and stores only task events, exported feature fields, and auxiliary context fields, without raw facial video persistence. In this sense, real-time browser-side processing is used to support feature-only local logging and timestamp alignment within the same client-side runtime, rather than to claim more accurate physiological estimates or online task adaptation.

To address these challenges this deployment scenario, we present **Camerale**, a privacy-conscious, browser-native framework designed for exploratory physiological sensing and task measurement across heterogeneous real-world environments for task-aligned physiological feature logging in remote task experiments.

1.1 Research Questions

We formulated two The exploratory Research Questions (RQs) to guide our deployment focus on task-synchronized face-derived records and descriptive sensor-derived observations under the tested configuration:

- **RQ1:** Can a browser-native, privacy-conscious sensing framework be operationally deployed across heterogeneous real-world environments without relying on external hardware or cloud video processing? In the intended remote psychological/psychophysiological task use case, do Camerale’s task-synchronized face-derived records, including rPPG-oriented ROI values, satisfy the operational criteria for descriptive analysis: trial-level reaction times within 100–1500 ms and exported-window-level landmark availability of $\text{mean_quality} \geq 0.8$ under the tested configuration?
- **RQ2:** What kinds of between-environment differences are observable in the physiological signals collected through such a deployment?

For RQ1, we evaluated the exported records using operational criteria at the trial and exported-window levels. At the trial level, trials were retained for descriptive analysis when reaction times fell within 100–1500 ms. At the exported-window level, landmark availability was evaluated using mean_quality , the mean of the binary landmark-availability flag within each 10-s window; exported windows with $\text{mean_quality} \geq 0.8$ were treated as satisfying the operational landmark-availability criterion. This criterion describes the exported frame/window records and should not be interpreted as face-detection availability across all camera frames. We also examined whether the exported records contained reproducible face-derived fields, including rPPG-oriented ROI values, motion estimates, and a frame-to-frame green-channel brightness-change indicator. In RQ2, the term “physiological signals” is used operationally to refer to task-synchronized face-derived and sensor-derived exported records, including rPPG-oriented ROI values and related exported fields. These records are not treated as validated physiological-state measurements. The present design was not able to determine whether laboratory/home differences reflected psychological

state, physiological state, lighting, screen light, posture, camera geometry, the measurement pipeline, task timing, participant state, or their mixture. Therefore, laboratory/home differences were interpreted as deployment-context differences rather than psychological, cognitive, motivational, or physiological effects. These analyses are intended only to describe operational completion and exported-record differences under the tested configuration; they do not demonstrate general deployability across users, devices, or environments.

1.2 Contributions

In this paper, we focus ~~strictly on the system architecture and its deployment feasibility. Specific contributions include~~ on system integration and conservative deployment observations. The contributions are:

1. **Privacy-Preserving Edge Architecture: A Local-first browser-native sensing pipeline that performs real-time rPPG and facial landmark extraction entirely on-device, avoiding raw video transmission.**
2. **System Design for Task-Aligned On-Device Sensing and Local Persistence: An implementation that keeps psychophysical system-integration implementation:** An implementation for remote psychological and psychophysiological task experiments that combines task execution, physiological signal extraction, and timestamped logging within the same client-side runtime, avoiding raw video export and server-side timing reconciliation ~~face-derived feature extraction, timestamp alignment, and local persistence within one browser runtime.~~
3. **Task-synchronized feature logging pipeline:** A logging pipeline that records task events, rPPG-oriented ROI values, FaceMesh-derived motion estimates, operational landmark-availability ratios, and frame-derived brightness-change indicators without raw facial video persistence.
4. **Exploratory In-the-Wild Deployment** deployment under limited conditions: A longitudinal case deployment across laboratory and home settings, showing that heterogeneous physical environments

~~materially influence deployment conditions and are associated with differences in observed signal characteristics.~~ An $N=1$ author-participant deployment showing that, under the tested configuration, the exported records satisfied the specified reaction-time and landmark-availability criteria for descriptive analysis.

The remainder of this paper is organized as follows:

Section II reviews ~~prior work in remote sensing and deployment-oriented data collection~~ related work in rPPG, web/edge physiological sensing, and task-measurement platforms. Section III ~~details the technical implementation of Camera~~ describes the Camera system architecture. Section IV describes the ~~exploratory deployment design and procedures~~ browser-based task deployment and reporting details. Section V presents descriptive ~~deployment observations and signal characteristics~~ observations. Section VI discusses implications for browser-native sensing design requirements and safeguards for future deployments, and Section VII concludes.

2. Related Work

2.1 Remote Physiological Sensing in the Wild Measurement and rPPG

~~Non-contact physiological measurement, particularly remote~~ Remote photoplethysmography (rPPG) ~~extracts pulse signals from facial micro-color variations. Methods range from classic signal processing like POS [2] to recent Deep Learning architectures [4]. While state-of-the-art networks offer high accuracy in laboratory settings, they frequently suffer from performance degradation across diverse physical environments when deployed in the wild under unstructured lighting or motion [5]. In our framework, we prioritize computational efficiency and robustness against rigid head motion, aiming to extract baselines rather than medical-grade clinical diagnostics.~~ estimates pulse-related color changes from facial video. Classic approaches include ambient-light remote plethysmographic imaging and blind-source video methods [1], [3], while more recent work includes deep-learning-based estimators [4]. These studies show that camera-based and webcam-based rPPG functionality already

exists. They also show that uncontrolled capture conditions, implementation choices, and evaluation pipelines are important factors in remote heart-rate imaging. Accordingly, Camerala does not claim to introduce a new rPPG estimator or to solve rPPG degradation under uncontrolled conditions. It records rPPG-oriented ROI values and exported indicators so that task-aligned records can be inspected descriptively.

2.2 ~~Systems for Task Measurement Browser-Based, Web-Deployable, Edge-Based, and Physiological Data Collection~~ Privacy-Oriented rPPG Systems

The shift toward remote and ubiquitous computing has generated several platforms for data collection. However, current systems face trade-offs between execution environment, privacy, and synchronized sensing. We map this design trade-off space in Table 1. Several closely related prior studies and systems are directly relevant to Camerala’s positioning. Ayesha *et al.* proposed a web application for experimenting with and validating remote measurement of vital signs [6]. LabVanced provides remote heart-rate detection/rPPG functionality within an online experiment platform [7]. FacePhys-Demo demonstrates browser-based rPPG monitoring with local inference in a public web demo/repository [8]. RhythmEdge addresses contactless heart-rate estimation on edge devices [10]. Gupta and Etemad investigated privacy-preserving remote heart-rate estimation from facial videos [11]. Di Lernia *et al.* evaluated rPPG via online webcams under uncontrolled conditions [12]. Together, these related studies and public systems indicate that browser-based, web-deployable, edge-based, and privacy-oriented rPPG have already been explored. Camerala therefore does not claim novelty in browser-based rPPG functionality, local rPPG processing, edge-oriented heart-rate estimation, or privacy preservation as an algorithm. Instead, Camerala is positioned as a system-integration implementation for remote psychological and psychophysiological task experiments: it combines browser-based task execution, face-derived feature extraction, task-event/frame-record timestamp alignment, operational landmark-availability

checking, and local CSV/ZIP persistence without raw facial video retention within one browser runtime.

~~Design Trade-off Mapping of Data Collection Platforms~~

2.3 Task Measurement and Physiological Data Collection Platforms

Well-established ~~behavioral experiment libraries~~ like jsPsych provide precise web experiment libraries and experiment runtimes, such as jsPsych [14] and PsychoPy/PsychoJS/Pavlovio [15], support browser-based task timing but do not natively integrate continuous webcam-based physiological sensing execution and online experiments. However, they do not generally provide an integrated webcam-derived physiological feature pipeline, task-sensing synchronization, and local-first feature logging as a single browser runtime. Conversely, rich Cloud sensing APIs provide deep analytics but fundamentally cloud analytics APIs can provide rich video-based analysis but often require transmitting video streams away from the client device, raising privacy barriers for domestic use. Camerala is designed to occupy a specific niche: merging the local execution of behavioral task timing with on-device physiological extraction (rPPG and facial landmarks), creating a unified runtime for exploratory “in-the-wild” deployments, participant’s device. Table 1 maps this design trade-off space.

Our contribution is not a new physiological estimator, but a browser-native integration that makes task-aligned, privacy-conscious physiological sensing deployable in domestic environments.

We structure the initial evaluation as a single-case exploratory deployment. The deployment is not intended to estimate population-level psychological or physiological effects. Instead, it documents whether the integrated browser runtime can complete the planned browser-based inference without cloud offloading. This technical shift acts as the enabler for Camerala task logging procedure and what measurement-quality differences are observed in the tested laboratory and home contexts.

3. System Architecture: Camerala

Camerala (derived from “Camera” + “Laboratory”) is a research prototype framework designed

Table 1 Design Trade-off Mapping of Data Collection Platforms

System / Platform	On-device video processing	Task-sensing synchronization	Local-first persistence	Raw video export	Typical use
Web experiment libraries	N/A (task only)	Partial	Depends on setup	N/A	Behavioral tasks
Experiment runtimes	Depends on deployment	Yes for task events	Depends on setup	Depends	Lab/online experiments
Browser or web rPPG systems	Often possible	Usually not task-centered	Varies	Varies	Vital-sign estimation
Cloud sensing APIs	No	Low or asynchronous	No	Often required	Video analytics
CameraLa	Client-side feature extraction	Task events and frame features in logs one runtime	Local ZIP export of logs	Not used by CameraLa	Remote task studies

To evaluate sensing systems under real-world environmental variability, longitudinal deployment is required. We structure this initial assessment as a Single-Case Exploratory Deployment ($N=1$) [9]. This approach focuses entirely on observing within-subject, between-environment (e.g., Lab vs. Home) variance in sensor behavior across numerous trials, avoiding cross-participant phenotypic variation in the initial deployment.

Recent advances in WebAssembly (Wasm) and WebGPU have made high-performance computer vision possible in the browser. By compiling C++ libraries (OpenCV, MediaPipe) to Wasm, we can enable Note: The table summarizes design emphasis rather than categorical superiority. Capabilities depend on implementation and deployment configuration.

to merge tightly synchronized psychophysical task procedures with on-device physiological signal extraction. The positioned as a local-first, browser-native research implementation for remote psychological and psychophysiological task experiments. Its contribution is not a new physiological estimator in isolation, but a deployment-oriented integration that combines on-device video analysis, synchronized browser-based rPPG, local rPPG processing, or privacy preservation as an algorithm. Rather, it integrates task execution, and local-first persistence in a single browser-native face-derived feature extraction, task-event/frame-record timestamp alignment, operational landmark-availability checking, and local CSV/ZIP persistence without raw facial video retention within one browser runtime. Although the current deployment used a single psychophysical task to measure feasibility deployment in this paper used one Gabor-orientation task, the runtime isolates the stimulus presentation separates task logic from the sensing pipeline (as depicted by the distinct modules in Fig. 1). As a supporting example of this modularity beyond the primary task used in the deployment, we implemented a supplementary spatial-memory task using the same sensing and logging pipeline; an example configuration and resulting JSONL log are included in the Appendix so that other task procedures can be implemented without changing the feature-extraction loop.

While asynchronous cloud analytics are awkward for tightly synchronized task deployments, and browser-native task runtimes currently do not provide built-in physiological extraction, CameraLa bridges this gap via an Edge-Computing philosophy. The combination of these features enables exploratory household deployments under privacy constraints. This architectural choice is driven by two critical non-functional requirements: **Privacy Preservation** and **Low-Latency Operation**.

3.1 High-Level Architecture

The system enables a complete experimental loop entirely within browser-based task experimental loop inside the browser sandbox (see Fig. 1). The architecture consists of three coupled modules running on the main JavaScript thread and dedicated WebWorkers:

1. **The Task Engine:** Responsible for stimulus Stimulus rendering, trial logic, and reaction time measurement key responses, feedback, staircase updates, and reaction-time logging.
2. **The Vision Pipeline:** Responsible for ingesting the webcam stream, performing facial landmark inference, and extracting raw physiological signals Webcam ingestion, MediaPipe Face Mesh inference, rPPG-oriented ROI feature extraction, head-motion estimates, blink/EAR features, and frame-level quality indicators.
3. **The Data Manager:** Responsible for fusing the asynchronous streams (Vision Frame Time

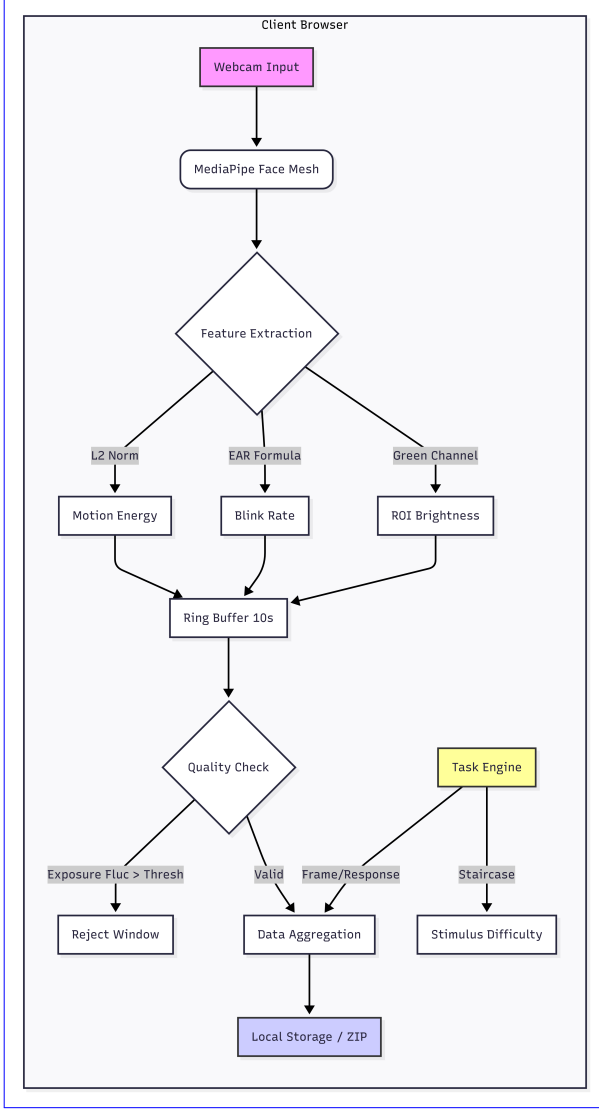


Fig.1 Overview of the Camera System Architecture. The system runs entirely client-side, processing browser obtains webcam frames via MediaPipe Face Mesh (Wasm) to extract rPPG, extracts FaceMesh/rPPG-oriented features and Head Motion signals in real-time. Data is synchronized quality indicators, synchronizes them with the Psychophysics Task Engine (Canvas API) task events, and stored locally exports local logs. Under the tested implementation, ensuring no only extracted features and task events are logged; raw video ever leaves the user's device is neither transmitted nor stored by Camera.

vs. Task Event Time) and securing data persistence Timestamp alignment, trial/frame aggregation, CSV serialization, ZIP packaging, and local persistence without raw-video storage.

3.2 Technical Implementation: The Vision Pipeline

We utilize the The current implementation uses MediaPipe Face Mesh [13] solution, which provides 468 3D facial landmarks in real-time. This model is executed via TensorFlow.js with the WebGL backend to leverage the client's GPU acceleration as a software framework for browser-based facial landmark extraction [16]. The camera stream is requested through getUserMedia with ideal constraints of 640×480 pixels and 30 fps. Because actual camera settings can depend on the browser and device, these constraints should be interpreted as requested settings rather than verified hardware settings.

3.2.1 Feature Extraction Logic

To quantify behaviorally relevant and motion-related features, we extract two primary signals: Head Motion Index and Exposure Fluctuation. The present implementation records sensor-derived features and quality indicators rather than validated clinical or medical measurements. Two categories are central to this paper:

1. Head Motion Index (M_t) To estimate participant stability and gross motor movement, we track the displacement of rigid cranial landmarks (Nose tip, Left/Right Tragus) that are generally unaffected by facial expressions. Let $L_i^t = (x_i^t, y_i^t, z_i^t)$ be the 3D coordinate of the i -th landmark. Head motion is approximated from the frame-to-frame displacement of a facial landmark. Let $L^t = (x^t, y^t)$ be the normalized landmark coordinate at frame t . The instantaneous motion scalar M_t is defined as the Euclidean distance summed over the set of rigid landmarks $S_{rigid} = \{1, 33, 263\}$ (Nose, L-Ear, R-Ear): motion scalar is computed as:

$$M_t = \frac{1}{|S_{rigid}|} \sum_{i \in S_{rigid}} \|L_i^t - L_i^{t-1}\|_2. \quad (1)$$

To ensure scale invariance (handling users sitting at different distances), the coordinates are normalized by the Inter-Ocular Distance (IOD), calculated as the Euclidean distance between the lateral canthi of the eyes. This feature is retained as a motion-related exported field for descriptive inspection of task-synchronized records. It is not an algorithmic correction for motion artifacts in rPPG and is not used to compensate for motion-induced rPPG degradation.

2. Exposure Fluctuation (E_{fluc}) ROI Green-Channel Value and Brightness-Change Indicator Deploying sensors outside the lab introduces variable lighting. In a home environment, clouds passing the sun, screens flickering, or power cycles induce luminance changes that can mask the rPPG signal. We quantify this frame-level luminance fluctuation as an environmental quality indicator. We define a Region of Interest (ROI) on the forehead, R_{fore} . The average luminance Y_t is computed from the RGB channels:

$$Y_t = \frac{1}{N_{px}} \sum_{(u,v) \in R_{fore}} (0.299R_{u,v} + 0.587G_{u,v} + 0.114B_{u,v})$$

The implementation samples a small forehead region around a FaceMesh landmark and records the green-channel brightness as an rPPG-oriented ROI value. It also computes a frame-to-frame green-channel brightness-change indicator:

$$B_{\Delta}(t) = |G_t - G_{t-1}|, \quad (2)$$

The Exposure Fluctuation metric is then defined as the first derivative of luminance:

$$E_{fluc}(t) = |Y_t - Y_{t-1}|$$

where G_t is the downsampled frame-level green-channel average at frame t . Camera1a is presented as a deployment and logging architecture, not as a method for improving rPPG signal quality under motion or illumination changes. Instead, it records motion-related and brightness-related exported fields for descriptive inspection of exported records. The brightness-change indicator does not measure camera exposure settings, external illuminance, specific environmental events, psychological or physiological state, or rPPG accuracy.

3.3 Technical Implementation: The Task Engine

The experimental psychophysical probe is a **2-Alternative Forced Choice (2AFC) Orientation Discrimination Task**.

3.3.1 Stimulus Generation via Canvas API

two-alternative forced-choice orientation discrimination task. Using the HTML5 Canvas API, we procedurally generate Gabor patches. A Gabor patch is defined as a sinusoidal grating multiplied by a Gaussian envelope. The luminance

profile $L(x, y)$ is given by:

$$L(x, y) = L_0 + C \cdot \exp\left(-\frac{x'^2 + \gamma^2 y'^2}{2\sigma^2}\right) \cdot \cos\left(2\pi \frac{x'}{\lambda} + \phi\right)$$

Where: L_0 : Background luminance (128/255 gray). C : Contrast of the grating. λ : Wavelength (spatial frequency). θ : Orientation angle (the target variable: $+45^\circ$ or -45°). x', y' : Coordinates rotated by θ . The task engine procedurally renders oriented grating stimuli, records key responses, calculates reaction time with `performance.now()`, and updates task difficulty. Procedural rendering avoids network-dependent stimulus loading during a trial and keeps task timing in the same browser runtime as feature logging. By generating stimuli procedurally rather than loading image files, we eliminate network latency and ensure pixel-perfect control over the spatial frequency and contrast.

3.3.1 Timing and Synchronization

Web-based reaction time measurement is notoriously jittery. To mitigate this, we employ The implementation uses the following timing strategy:

1. **High-Res Time High-resolution timestamps:** All timestamps use Task events and frame features are timestamped using `performance.now()`, which provides high-resolution timestamps independent of the system clock.
2. **Frame-Locked Rendering Frame-loop alignment:** Stimulus rendering is synchronized using `requestAnimationFrame`. We log the timestamp of the frame immediately before the stimulus paint command. Feature extraction and task rendering are coordinated with the browser frame loop.
3. **Low-Latency Sync Local runtime alignment:** On-device processing avoids network transmission during task-aligned sensing and keeps task and sensing timestamps within the same local runtime in the present design. Task events and sensor-derived features are logged in the same browser runtime, avoiding server-side reconciliation.

3.4 Data Management and Privacy Local-First Persistence

Since no server is involved, data persistence is handled via the **File System Access API** (on supported browsers) or a fallback to Blob download. Data is serialized into JSON Lines

Algorithm 1 Camerale Task Loop (Simplified)

```
Initialize Camera, FaceMesh, Canvas
StartContinuousVisionFrameLoop()
while Session Active do
  Block ← GetCurrentBlock()
  Condition ← Block.condition
  Context ← Block.instruction
  ShowInstruction(ConditionContext.text)
  for trial  $i = 1$  to  $n$  do
    Fixation Phase: Show
    Cross  $(500ms + \mathcal{U}(0, 200)ms)$   $\theta \leftarrow$ 
    SampleOrientation( $\{-45^\circ, +45^\circ\}$ )
    ShowFixation(400–600 ms)
     $t_{onset} \leftarrow$  performance.now()
    DrawGaborPatch( $\theta$ ; DrawOrientationStimulus( $\theta$ ,
    200 ms))
    while No KeyPress do
      Poll Keyboard Input F/J KeyPress
    end while
     $t_{response} \leftarrow$  performance.now()
     $RT \leftarrow t_{response} - t_{onset}$ 
    Accuracy ← (Key == Target) Feedback:
    Show Color (Green/Red) LogTaskEvent( $RT$ ,
    Accuracy, Context)
    UpdateStaircase(Accuracy, relative  $\pm 5\%$ 
    contrast step)
    Log( $RT$ , Accuracy, FaceMeshFeatures)
    LinkTaskEventToFrameTimestamps( $t_{onset}$ ,
     $t_{response}$ , Context)
  end for
end while
```

(.jsonl) format. Each line represents one frame or one trial event. This allows for streaming writes, ensuring that even if the browser crashes mid-session, data up to that point is preserved. At the end of a session The current implementation uses JSZip to export local log files as a ZIP archive. The archive contains `metadata.json`, `trials.csv`, `features_raw.csv`, the user initiates a "Download Data" action, which zips the JSON logs and saves them to their local disk. This explicit user action reinforces the sense of agency and privacy control: `windows.csv`, `blocks.csv`, and `subjective.csv`. The metadata file stores the session ID, start time, user agent, and block plan. The user initiates the download at the end of the session. This design avoids Camerale-side raw-video storage and avoids raw-video transmission, but it also reduces the ability to inspect image artifacts retrospectively. For this reason, Camerale logs reproducible exported fields such as rPPG-oriented ROI green-channel values, head-motion estimates, and a frame-to-frame green-channel brightness-change indicator.

4. Methodology

4.1 Longitudinal Target Usage Scenario and Exploratory Deployment To evaluate the operational deployability of Camerale across different physical environments

The deployment was a browser-based task experiment conducted in one laboratory setting and one home setting. The participant sat in front of a laptop, viewed Gabor stimuli, and responded using the keyboard. This setting should be distinguished from unconstrained daily-life physiological sensing during walking, conversation, cooking, sleep, or other ordinary activities.

To evaluate operational completion under this scenario, we conducted an exploratory longitudinal deployment ($N=1$). While traditional between-subjects group designs (Large- N) are strictly necessary for estimating generalizable psychological effects, an intensive $N=1$ autoethnography (Trials=with one author-participant. The deployment consisted of 10 sessions and 600) serves as a technical proof-of-concept for the system architecture. This design allows us to observe the system's operational

continuity across contrasting ecological conditions without the compounding variance of diverse user demographics. The study followed an A-B-C logic within each session (see Fig. 4), varying the explicit instructional framing conditions presented to the user (Neutral, Negative Evaluation, Positive Evaluation). Each session consisted of 3 Blocks (60 trials total), order-balanced via a Latin Square design to prevent order effects total trials. Each session consisted of six 10-trial blocks, with two blocks for each instructional task context (neutral instruction, negative instruction, and positive instruction), as reflected in the exported block plans. The order of blocks was randomized by the web application.

Visual representation of the deployment schedule. The study spanned 14 days, alternating between Home and Lab environments. Each session consisted of 3 blocks with randomized order, totaling 600 trials. This approach allows for robust observation of between-environment signal differences.

4.2 Participant (Autoethnography)

In line with standard systems feasibility evaluation, the The participant was the primary author. *Ethical Note:* This was a self-experimentation (autoethnographic) evaluation. No evaluation: no other participants were involved, no identifiable raw video was transmitted or shared by Cameralea, and the author provided informed consent for self-participation. We explicitly acknowledge that the author's prior knowledge of the experimental hypotheses precludes drawing population-level psychological conclusions. However, this deployment is methodologically appropriate for the initial technical validation of the sensing pipeline. Because the participant was also the author and knew that the task instructions were experimentally constructed, the instruction blocks cannot be treated as validated manipulations of evaluation pressure, reward motivation, cognitive state, motivational state, or psychological state. They are used only as exploratory task-interface contexts for demonstrating synchronized logging across task conditions.

4.3 Apparatus and Physical Environment

The experiment was deployed on a standard 13.5-inch laptop display (approximate 60Hz target refresh rate). To contextualize the environmental

differences, ambient illuminance was estimated using a digital lux meter at the eye level of the participant.

Lab Environment: Mean illuminance 480 ± 20 lux (Standard fluorescent lighting).

4.3 Hardware and Software Configuration

Table 2 summarizes the hardware and software information available for the deployment. Values that were not logged during the deployment are explicitly marked as not logged rather than inferred from the revision-time environment.

4.4 Deployment Environment and Lighting Conditions

The deployment was conducted during daytime in an indoor university laboratory/research room and an indoor private home room under general artificial room lighting. Both rooms had windows, but curtains were closed during the sessions. Direct natural light was not used as the primary illumination source in either context. The built-in laptop display brightness was fixed across sessions. Because the deployment used camera-derived setup checks rather than external light-meter measurements, this paper reports brightness ranges only as author-confirmed setup/check information. The camera-derived brightness range observed during setup/checks was 105–114 in the laboratory setting and 102–111 in the home setting. Table 3 summarizes the deployment environment and lighting conditions.

4.5 Experimental Procedure & Cognitive Framing and Instructional Task Contexts

Each session began with an automated a pre-session check used to screen basic capture capture check for basic camera conditions, including luminance, pose, and face size prior to hiding approximate camera-derived brightness and pose, before the camera feed to prevent distraction. We then presented distinct instructional framing conditions (Neutral, Negative Evaluation, Positive Evaluation) via full-screen text before each 20-trial block. The **Neutral** condition served as a baseline. The **Negative Evaluation** condition ("WARNING: CRITICAL. Poor performance is unacceptable") served as a negative task framing manipulation. The **Positive Evaluation** condition ("BONUS. Break your record!") served as a positive task framing manipulation was hidden to reduce

Table 2 Hardware and software configuration

Item	Reported value
Laptop/PC model	mouse Computer G-Tune P517G60ZO21CNHWT
CPU	Intel Core i7-13620H, 10 cores (P6+E4), 16 threads
GPU	NVIDIA GeForce RTX 5060 Laptop GPU
RAM	32 GB
Storage	1 TB SSD
Operating system	Windows 11 Home 64-bit; exact deployment-time build not logged
Display	15.6-inch display, 2560×1440 pixels, 165 Hz
Browsers used	Mozilla Firefox and Google Chrome
Browser versions	Firefox user-agent was available when recorded in exported metadata; exact Chrome deployment-time version was not systematically logged
Webcam type	Built-in laptop webcam
Webcam model	Exact internal webcam model not logged
Camera capture API	Browser <code>getUserMedia</code> API
Requested camera constraints	Ideal 640×480 pixels, ideal 30 fps
Actual camera resolution	Browser/device-negotiated; not programmatically logged as a verified deployment-time value
Actual camera frame rate	Browser/device-negotiated; not programmatically logged as a verified camera frame rate
Screen brightness	Fixed during the deployment; exact percentage not logged
Auto-exposure	Device/browser default behavior; not manually constrained and exact state not programmatically logged
Auto-white-balance	Device/browser default behavior; not manually constrained and exact state not programmatically logged

distraction. The task included negative, neutral, and positive instruction contexts as exploratory task-interface conditions. These contexts were used to demonstrate that Cameralea can present different instruction frames while recording synchronized behavioral and face-derived records. The labels “negative” and “positive” refer only to instruction labels used in the task interface. Because the participant was the author and knew that the instructions were experimentally constructed, these contexts are not treated as validated manipulations of evaluation pressure, reward motivation, cognitive state, or motivational state.

4.6 The Psychophysical Task

The psychophysical probe was a Gabor Orientation Discrimination task. The task was a two-alternative forced-choice orientation discrimination task using procedurally rendered Gabor stimuli. Each trial began with

a randomized fixation cross (400–600ms), a brief Gabor stimulus ($\pm 45^\circ$, 200ms), a response window, and immediate visual feedback period of 400–600 ms, followed by a Gabor stimulus oriented at either $+45^\circ$ or -45° for 200 ms. The participant responded using the F/J keys: the F key corresponded to -45° and the J key corresponded to $+45^\circ$. The application recorded stimulus onset, response key, reaction time, correctness, and task difficulty together with frame-level sensor-derived features. To maintain a consistent level of adaptive task difficulty, we implemented the implementation used a standard 1-up 1-down staircase algorithm procedure on stimulus contrast. By adjusting contrast dynamically ($\pm 5\%$ relative steps) based on Stimulus contrast was adjusted by relative $\pm 5\%$ steps according to the previous response, the task adaptively targeted the participant’s 50% psychometric threshold across all sessions. These task parameters are reported

Table 3 Deployment environment and lighting conditions

Item	Laboratory	HomeEnvironment :—Mean illuminance 140 ± 50 lux (Mixed LED and variable monitor glow). Distance from the screen was approximately maintained at 50–60 cm, using the FaceMesh tracking standard distance as a rough proxy.—
Room type	Indoor university laboratory/research room	Indoor private home room
Time of day	Daytime	Daytime
Primary lighting	General indoor artificial lighting	General indoor artificial lighting
Window presence	Windows present	Windows present
Curtain/blind condition	Curtains present and closed	Curtains present and closed
Natural light contribution	Direct natural light was not used as the primary illumination source	Direct natural light was not used as the primary illumination source
Camera-derived brightness range observed during setup/checks	105–114	102–111
External light-meter measurement	External light-meter values were not used; the reported ranges are camera-derived setup/check information, not lux measurements.	
Screen brightness	Fixed across sessions using the built-in laptop display setting.	
Task display	Stimuli were presented on the built-in laptop display. Screen luminance was part of the tested configuration rather than an independently manipulated factor.	
Window aggregation	Exported window-level features used 10-s windows with 5-s overlap.	
Brightness-change indicator aggregation	The frame-to-frame green-channel brightness-change indicator was summarized in 10-s windows with 5-s overlap.	
Clouds/screen flicker/power cycles	Not treated as observed experimental events; possible brightness variation was handled only as camera-image-derived record variation.	

for reproducibility; the instruction blocks are not interpreted as validated cognitive or motivational manipulations.

4.7 Dependent Measures and Pre-processing

All ~~data were exported as JSONL and~~ exported logs were processed sequentially. The primary behavioral measure was reaction time (RT), with an operational analysis range of 100–1500 ms. The primary sensor-derived measures were rPPG-oriented ROI green-channel values, head-motion estimates, and the frame-to-frame green-channel brightness-change indicator. Behavioral trials were retained when reaction times fell within 100–1500 ms. Cameralea used MediaPipe FaceMesh to obtain facial landmarks in the browser. Frame records forwarded from the FaceMesh callback were synchronized with

task events during active sessions. The feature extractor defines a binary landmark-availability flag, `quality`, where 1 indicates that FaceMesh landmarks were returned and 0 indicates that no landmarks were returned. In the current runtime, however, the camera manager forwards results to the logging callback only when FaceMesh landmarks are returned; therefore, the saved frame records in this deployment represent landmark-returned exported frames. For each exported 10-s window, `mean_quality` was computed as the mean of the `quality` flag over the exported frame records in that window. Thus, `mean_quality` describes landmark availability within exported frame/window records and not face-detection availability across all camera frames. This value is not a continuous MediaPipe confidence score, an indicator of rPPG measurement quality, or evidence of physiological measurement

accuracy. The FaceMesh `minDetectionConfidence` and `minTrackingConfidence` parameters were both set to 0.5 as implementation settings. They were not used as post-hoc analysis thresholds.

4.7.1 Task-Performance Metrics

- **Reaction Time (RT):** Limited to an operational range (100ms to 1500ms) to exclude task distractions.
- **Accuracy:** Tracked to verify staircase convergence.

4.7.1 Physiological Metrics

- **Head Motion Index (M_t):** Calculated as the frame-by-frame Euclidean displacement of rigid facial landmarks.
- **rPPG Heart Rate:** The raw POS signal trace was bandpass filtered (0.7-3.0 Hz). Peak detection was applied to estimate inter-beat intervals (IBI).

5. Results

Our primary objective in this section is to evaluate the operational feasibility and signal quality of the framework across heterogeneous real-world settings: a controlled University Laboratory versus an uncontrolled private Home environment. Data from The results are reported descriptively as sensor-derived observations under the tested configuration. The deployment completed all 10 sessions (30 blocks, planned sessions and 600 trials in total. After applying the reaction-time criterion of 100-1500 ms, 541/600 trials) were successfully collected and persisted entirely via the local browser framework. We filtered trials with extreme reaction times ($RT < 100ms$ or $RT > 1500ms$) and low face tracking confidence ($Conf < 0.8$), resulting in 582 valid trials (97.0% retention). were retained for descriptive analysis. The retained trials were 256/300 in the laboratory setting and 285/300 in the home setting. All 299 exported windows had `mean_quality = 1.0` and therefore satisfied the operational landmark-availability criterion of `mean_quality  $\geq$  0.8`; no window was excluded by this criterion. These results describe the available analyzed trials and exported windows under the tested configuration and do not demonstrate general feasibility across devices, users, or environments.

5.1 Environmental Noise and Signal Extraction Pipeline

Before observing behavioral differences, we assessed the stability of

5.1 Task and Exported Feature Summaries

Table 4 summarizes task-level and exported feature summaries in the laboratory and home contexts. After RT exclusion, mean RT was 721.1 ± 257.7 ms in the laboratory setting and 676.5 ± 249.5 ms in the client-side rPPG and FaceMesh pipeline across the two environments. As shown in Table 4 and Fig. 2, the Home environment presented greater challenges for signal quality. home setting. Accuracy was 85.5% and 82.1%, respectively. Window-level exported records showed descriptive differences in ROI green-channel values, motion estimates, and the brightness-change indicator between settings. Because lighting, screen light, posture, camera geometry, task timing, and participant state were not independently controlled, these differences are reported only as deployment-context differences in exported records.

Data Quality Comparison. The Home environment (Orange) exhibits higher exposure fluctuation (E_{fluc}) and lower tracking confidence compared to the Laboratory (Blue), posing a challenge for rPPG signal extraction.

Descriptive Data Quality Comparison (Trial-Level Aggregation)

The "Exposure Fluctuation" (E_{fluc}) was noticeably higher in the Home environment on average, indicating a difference in environmental lighting artifacts. Despite this, the tracking confidence remained above 96% at home, supporting the feasibility of using the MediaPipe + WebAssembly stack for this naturalistic deployment.

5.2 Observation of Task Performance Differences

To investigate

5.2 Task-Event and Sensor-Derived Observations

To address RQ2 (between-environment differences), we aggregated Reaction Time (RT) across the three instructional framing conditions (Neutral, Negative Evaluation, Positive Evaluation) in both the Laboratory and Home environments. We summarize trial-level variation descriptively to examine whether coarse between-environment

Table 4 Descriptive task and exported feature summaries after reaction-time exclusion.

Metric	Laboratory	Home	Notes
Retained trials after RT exclusion	256/300 (85.3%)	285/300 (95.0%)	Trial-level; RT 100–1500 ms
Mean RT after RT exclusion	721.1 $\pm$ 257.7 ms	676.5 $\pm$ 249.5 ms	Trial-level retained trials
Accuracy after RT exclusion	85.5%	82.1%	Trial-level retained trials
rPPG-oriented ROI green-channel value	46.69 $\pm$ 6.80	34.32 $\pm$ 2.72	Window-level; windows.csv
Motion estimate	0.0011 $\pm$ 0.0005	0.0010 $\pm$ 0.0004	Window-level; windows.csv
Frame-to-frame green-channel brightness-change indicator	0.0504 $\pm$ 0.0509	0.0707 $\pm$ 0.0337	Window-level; windows.csv

differences were observable in this exploratory dataset.

Descriptive trends in Reaction Time (RT) grouped by framing condition. Under controlled Laboratory settings, the negative instructional framing exhibited descriptive deviations from the Neutral condition. At Home, variance across conditions appeared to mask task-induced deviations. Error bars represent Standard Error (SE).

5.2.1 Laboratory Environment

In the controlled Lab environment, the Negative Evaluation condition exhibited an observable acceleration in mean responses ($M = 485ms$, $SD = 82$) relative to the Neutral condition ($M = 520ms$, $SD = 95$). we compared descriptive task-synchronized records between the laboratory and home contexts. The exported records showed differences in rPPG-oriented ROI green-channel values, motion estimates, and the frame-to-frame green-channel brightness-change indicator. Because lighting, screen light, posture, camera geometry, task timing, and participant state were not independently controlled, these differences are reported only as deployment-context differences in exported records. They should not be interpreted as evidence that the instruction blocks induced psychological, cognitive, motivational, or physiological effects. Motion estimates are reported as exported face-derived records for descriptive comparison between the laboratory and home contexts. They are not interpreted as evidence of motion handling or correction of rPPG degradation. This pattern indicates that the task-derived response metric was more clearly separated from the neutral condition in the laboratory setting.

5.2.1 Home Environment

Conversely, in the Home environment, RTs across all conditions clustered closely around the Neutral condition (Negative: $M = 512ms$; Neutral: $M = 515ms$). The descriptive separation observed in the laboratory setting was not apparent in the home setting. This highlights the susceptibility of specific task-derived metrics to masking by uncontrolled domestic noise factors. The instruction contexts were used only as task-interface contexts in the deployment. We do not report them as evidence that evaluation pressure, reward motivation, cognitive effects, or motivational effects were induced. Descriptive values are reported by instruction context only to document that the logging pipeline operated across different task-interface contexts.

5.3 Between-Environment Differences in Physiological Features

We further investigated if differences extended to the collected physiological estimates.

5.2.1 Head Motion

We analyzed the descriptive Head Motion Index (M_t) during the instruction-reading phase. In the Lab, the Negative Evaluation condition showed a reduction in head motion compared to the Neutral condition. At Home, however, higher motion variability was observed during the same instruction. laboratory context, the negative instruction block showed a lower mean RT than the neutral block in this dataset. In the home context, the mean RT values across instruction blocks were closer together. Because the deployment used one author-participant and did not isolate instruction effects from environment, lighting, screen light, posture, camera geometry, or measurement-pipeline factors, these RT patterns are reported only as descriptive task-event observations.

5.2.1 Heart Rate Analysis (rPPG)

Due to the short trial duration (seconds per trial), we aggregated HR estimation across the entire block duration (approx. 2 minutes). While the rPPG signal quality in the Lab provided clean peaks, the Home data suffered from noise artifacts during high-motion segments. After excluding segments with high optical motion noise ($M_t > 2.0$), the observed

5.3 rPPG-Oriented ROI Estimates

The exported data include rPPG-oriented ROI brightness values and related quality indicators, but no ECG or contact PPG reference was collected. Therefore, this paper does not interpret ROI-derived or block-level descriptive differences were:

- **Lab:** The Negative Evaluation condition showed an average HR elevation of approximately +8.5 bpm relative to the Neutral condition.
- **Home:** The same condition showed a mean HR change of +1.2 bpm.

These physiological observations were reported alongside the task-derived trends and showed descriptive differences across environments summaries as validated heart-rate changes. The appropriate interpretation is that the tested deployment yielded face-derived and sensor-derived exported records whose values differed descriptively between the tested laboratory and home contexts.

6. Discussion

The central finding/main implication of this deployment is not a population-level effect of instructional framing, but that identical task procedures produced different observed task and physiological patterns across environments. For system evaluation, this suggests that laboratory-only validation may overestimate signal stability and condition separability under domestic use. This highlights an engineering constraint for continuous sensing: systems evaluated purely on Lab data may exhibit environment-associated changes in observed signals and measurement quality when moved from laboratory to domestic settings.

6.1 Prioritizing Browser-Native Architectures

Methodologically, we demonstrated that a that home or laboratory settings affect camera-based physiological records; this is a known limitation of video-based sensing. Instead, the deployment illustrates design requirements for a local-first, browser-native framework (CameraLa) can operationally collect synchronized behavioral and physiological data in domestic spaces without transmitting raw video. By retaining computation purely on the client side, the architecture minimizes the privacy barriers that often inhibit exploratory "in-the-wild" deployments involving video sensors. logging architecture for remote psychological and psychophysiological task experiments. In particular, when raw facial video is not retained, the system must preserve task-event timestamps, exported face-derived records, operational landmark-availability checks, motion and brightness-change records, and sufficient hardware/environment metadata to support cautious post-hoc interpretation. Under the tested configuration, CameraLa completed the planned task procedure and exported task-synchronized records for descriptive analysis. However, the laboratory/home differences should be interpreted as mixed deployment-context differences rather than psychological, cognitive, motivational, or physiological effects. Future versions should include pre-session camera and lighting checks, explicit operational criteria, reference physiological sensors, multiple participants, multiple devices, and controlled lighting, posture, and camera conditions.

CameraLa's useful system property is integration: task timing, frame-level feature extraction, quality indicators, and local export are handled in one client-side workflow. This can reduce raw-video handling and server-side timestamp reconciliation, but it also removes the possibility of visually inspecting many artifacts after data collection. The trade-off makes quality-aware logging important. The relevant indicators include landmark availability, rPPG-oriented ROI values, head-motion estimates, and camera-image-derived brightness-change records; these values can support pre-session checks and descriptive inspection, but they are not proof that rPPG estimates are accurate or that motion-induced rPPG degradation has been

corrected.

The laboratory/home differences observed in this deployment should be interpreted as mixed deployment-context differences that may reflect environmental conditions, screen light, posture, camera geometry, the measurement pipeline, task timing, participant state, or their mixture. The study was not designed to isolate psychological or physiological state changes from these factors, and it cannot validate the psychological effectiveness of the negative or positive instruction contexts. Future deployments should therefore incorporate task-event/frame-record timestamp alignment, operational landmark-availability criteria, documented hardware/browser/camera settings, explicit reporting of camera constraints and actual camera settings, fixed screen-brightness settings, pre-session camera/lighting checks, landmark-availability warnings, motion and camera-derived brightness-change records, environment reporting, and specified quality-aware analysis rules. If privacy-sensitive raw video is not stored, these safeguards become more important because many artifacts cannot be retrospectively inspected from images; for studies that aim to make claims about physiological responses, a reference physiological sensor and a multi-participant design are necessary.

6.1 Limitations

There are several strict limitations to this work. First, this is a single-case exploratory deployment (This exploratory N=1); the results represent observational, deployment-oriented feasibility evidence rather than generalizable population inference. Second, the observed between-environment differences in physiological signals do not definitively prove cognitive state changes; they may be heavily confounded by uncontrolled physical variables including ambient lighting changes, subtle posture shifts, display luminescence, or camera placement artifacts inherent to unstructured deployments. Furthermore, the current work does not establish comparative superiority over existing runtimes or APIs. Future large-scale deployments must attempt to disentangle physical artifacts from cognitive shifts through sensor-fused studies. Even with these limitations, the present study establishes a concrete

author-participant deployment used one primary laptop/camera configuration, no ECG or contact PPG reference, and no comparison with existing experiment runtimes, browser rPPG tools, or cloud APIs. It did not test multiple participants, devices, or home environments, and it did not experimentally separate lighting, screen light, posture, camera placement, task timing, participant state, and task context. The implementation does not correct motion-induced rPPG degradation, and the logs do not store continuous confidence values for all frames; therefore, failed landmark-detection frames and overall tracking-success rates cannot be recovered. Future work should use reference physiological sensors, controlled posture/camera conditions, predefined artifact-handling rules, and continuous confidence logging with specified operational thresholds. The study does not claim clinical or medical accuracy, general deployability, general feasibility, or validated cognitive, motivational, psychological, or physiological effects; it presents an implementation example of browser-native ~~design pattern for privacy-conscious~~, task-aligned ~~sensing under heterogeneous physical~~, local-first feature logging under limited laboratory and home conditions.

7. Conclusion

~~Continuous physiological sensing in naturalistic environments requires tools that balance operational stability with stringent privacy protections. We presented CameraLa, a privacy-conscious, browser-native framework that extracts rPPG and facial landmarks on-device. Our proof-of-concept deployment provided initial evidence of the framework's browser-native architecture and highlighted the degree to which heterogeneous physical environments are associated with changes in observed task and physiological signals. These findings underscore the importance of evaluating sensor pipelines under domestic conditions and establish a concrete browser-native design pattern for synchronized, local-first implementation example that integrates psychophysical task execution, face-derived feature extraction, task-event/frame-record timestamp alignment, operational landmark-availability checking, and local CSV/ZIP persistence without raw~~

facial video retention. In a limited 10-session N=1 author-participant deployment across one laboratory setting and one home setting, Cameralea completed 600 trials, and 541/600 trials remained after the specified reaction-time criterion. The deployment produced synchronized behavioral logs and face-derived and sensor-derived records with descriptive deployment-context differences between the tested environments. These observations should not be interpreted as evidence of general deployability or physiological/cognitive effects; instead, they highlight design safeguards needed for future local-first physiological sensing under heterogeneous physical environments feature logging in home-based psychophysical experiments.

Data availability The dataset containing facial landmarks, extracted rPPG signals, and behavioral responses associated with this study

Data availability. The dataset cannot be made publicly available due to privacy restrictions and potentially identifiable biological data originating from domestic (in-the-wild) recordings because it contains privacy-sensitive face-derived records from a domestic deployment context.

CRedit authorship contribution statement

CRedit authorship contribution statement.

Kotaro Furukawa: Conceptualization, Methodology, Software, Investigation, Formal analysis, Writing - original draft.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper

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Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work the author(s) used Google Gemini in order to refine the LaTeX

code and format it according to the journal guidelines. After using this tool/service, the author(s) reviewed and edited the content as needed and take(s) full responsibility for the content of the published article.

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Supporting Example: Task Substitution

A core engineering feature of the framework is the separation between task logic and the physiological extraction engine. By standardizing the event logging interval around the `window.requestAnimationFrame` target cycle, the framework is designed so that cognitive task logic can be replaced without modifying the MediaPipe vision thread. **Listing 1** illustrates a pseudocode configuration for substituting the Gabor orientation task with a generalized Spatial Memory task block.

```
const task = {  
  stimulus: renderSpatialGrid(memory_load_target),  
  allowed_keys: ["Space"],  
  on_frame_sync: (timestamp_ms) => {  
    camera_logger.logEvent("stimulus_onset", timestamp_ms);  
  },  
};
```

To provide a supporting example of this separation, the structure ensures that time-critical physiological measurements continue asynchronously, while semantic trial events (e.g., *stimulus_onset*) are dynamically injected into the timeline. **Listing 2** shows the resulting unified data stream where event markers coexist with dense sensor vectors.

```
{ "time_ms": 4015.2, "event": "grid_on", "load": 4, "rppg_g": 110.4, "mot_x": 0.02 }  
{ "time_ms": 4016.8, "event": "frame_sync", "load": 4, "rppg_g": 109.8, "mot_x": 0.02 }  
{ "time_ms": 4032.6, "event": "frame_sync", "load": 4, "rppg_g": 111.1, "mot_x": 0.01 }  
{ "time_ms": 4211.4, "event": "response", "key": "Space", "rt_ms": 196.2, "rppg_g": 112.6 }
```